

Explainable Fuzzy Deep Learning for Prediction of Epileptic Seizures Using EEG

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Abstract—Addressing the challenge posed by the unpredictable and recurrent nature of epileptic seizures, which stand among the most significant neurological conditions, remains imperative, especially within settings inundated with high patient flow. The prompt identification of these seizures is paramount for effective patient care. Unfortunately, existing epilepsy seizure detection systems encounter limitations in availability and interpretability, thereby constraining their reliability and widespread application. Presently, neurophysiologists heavily rely on visually interpreting electroencephalogram (EEG) recordings displayed on screens to identify seizures. This article introduces an innovative method dedicated to detecting epileptic seizures within EEG signals, leveraging a specifically tailored fuzzy deep learning (FDL) architecture. The proposed methodology encompasses crucial stages of preprocessing and feature extraction, augmented by the utilization of explainable artificial intelligence models, such as local interpretable model-agnostic explanations (LIME) and Shapley additive explanation (SHAP) for enhancing model interpretability. The developed FDL model demonstrates promising results, achieving a noteworthy accuracy of 92.57%, precision of 0.96 for “normal” and 0.89 for “abnormal,” recall of 0.91 for “normal” and 0.94 for “abnormal,” and F1-score of 0.93 for “normal” and 0.91 for “abnormal,” affirming its robustness in classification tasks. In addition, to validate the effectiveness of the proposed FDL, comparisons are performed with long-short term memory networks and 1-D convolutional neural network model models. The integration of LIME and SHAP significantly enhances the interpretability of the model, providing valuable insights into influential features. This comprehensive framework adeptly balances accuracy and interpretability, thereby making a substantial stride in advancing EEG-based diagnostic tools.

Index Terms—Classification, epileptic seizure, explainable artificial intelligence (XAI), fuzzy deep learning (FDL).

I. INTRODUCTION

EPILEPSY stands as a neurological disorder characterized by recurrent and unpredictable seizures, impacting millions of individuals globally [1]. Leveraging the richness of time,

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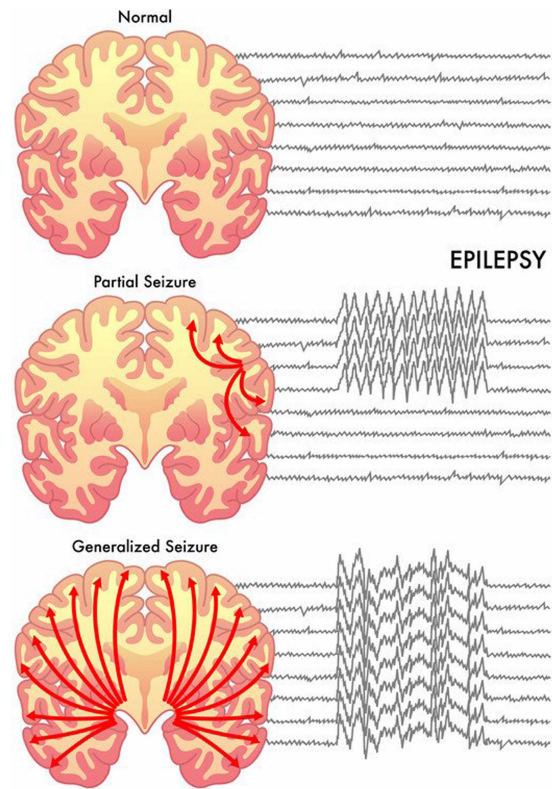


Fig. 1. EEG patterns for normal and epilepsy cases.

frequency, and spatial information embedded in nonstationary and transient electroencephalography signals through advanced machine learning (ML) and deep learning (DL) algorithms unlocks exciting possibilities for brain–computer interfacing [2] and medical diagnosis. EEG has emerged as a pivotal tool in both diagnosing and monitoring epilepsy, offering real-time insights into brain activity. The EEG patterns for normal, partial, and generalized epilepsy are different, as shown in Fig. 1. The accurate and prompt identification of epileptic seizures from EEG signals remains critical for timely medical interventions and enhanced patient outcomes [3]. Conventional methods for detecting epileptic seizures have traditionally relied on signal processing techniques and ML algorithms. However, the intricate and nonlinear nature of EEG data presents challenges for these conventional approaches. In recent years, the advent of DL has surfaced as a potent tool across various domains, showcasing

exceptional success in tasks, such as image recognition, natural language processing, and medical image analysis. Within the domain of epilepsy detection, DL models—particularly deep neural networks—have exhibited promise in automatically discerning intricate patterns from EEG data. However, the inherent closed-box nature of these DL models raises concerns regarding their interpretability, thereby limiting their applicability in critical medical domains where transparency and explainability are imperative.

In response to this challenge, our research aims to integrate explainable fuzzy logic with DL methodologies for epileptic seizure detection. Fuzzy logic provides an intuitive basis for representing uncertainty and inaccuracy, aiding in the clearer interpretation of complex connections within EEG data. Through the fusion of the robust capabilities of DL and the transparent nature of fuzzy logic, our proposed approach strives to augment the reliability of seizure detection models. While DL has shown remarkable performance in various applications, the lack of transparency in these models hinders their acceptance in medical settings, where understanding the decision-making process is paramount. By incorporating fuzzy logic, which inherently captures uncertainty and vagueness, our goal is to offer a clearer and more understandable framework for analyzing the complex patterns linked to epileptic seizures in EEG data. This concerted effort is aimed at rendering these models more accessible and comprehensible for clinicians and healthcare practitioners.

Our study employs a comprehensive methodology that integrates preprocessing, feature extraction, model architecture, and explainable artificial intelligence (XAI) techniques. In the preprocessing phase, EEG signals are cleaned and prepared for analysis. Feature extraction involves extracting a range of statistical features. The model architecture is designed to leverage both long short-term memory networks (LSTM) and 1-D convolutional neural network (1D-CNN) independently to benchmark against our proposed fuzzy deep learning (FDL) model. This comparative analysis highlights the strengths and weaknesses of each approach in capturing the temporal and local patterns in EEG signals.

Sections III-E and III-F detail the LSTM and 1D-CNN architectures, respectively, which were independently deployed to compare their performance with the proposed FDL model. The integration process is designed to ensure that the preprocessing and feature extraction stages provide the necessary input for all three models, facilitating a seamless comparison. The FDL model integrates fuzzy logic to enhance interpretability, addressing the closed-box nature of traditional DL models.

The rest of this article is organized as follows. Section II offers a comprehensive literature review of modern epileptic seizure detection techniques, highlighting the specific challenges addressed by our approach. Section III explores the methodology, encompassing details about the dataset, preprocessing techniques, feature extraction methods, and the architectural design of our explainable FDL model, which integrates principles derived from fuzzy logic. Sections III-E and III-F specifically discuss the LSTM and 1D-CNN models used for performance benchmarking. Section IV presents a detailed analysis of results, discussing experimental findings and emphasizing both

the interpretability and accuracy of the model. Finally, Section V concludes this article by outlining future directions in this field.

II. LITERATURE REVIEW

Recent years have witnessed significant advancements in automated seizure detection methods, utilizing EEG signals as a core source of information. This review aims to synthesize key findings and emerging trends in feature extraction, classification models, and future directions for this crucial research area.

Feature extraction plays a pivotal role in extracting relevant information from complex EEG data. Wavelet-based methods, such as discrete wavelet transform and variational modal decomposition, excel at capturing time–frequency information, particularly valuable for nonstationary EEG signals [4], [5]. Channel selection techniques, such as sparse common spatial pattern, reduce dimensionality while enhancing classifier performance by identifying informative channels [6]. DL techniques, particularly convolutional neural networks (CNNs), have emerged as powerful tools for directly extracting intricate features from raw EEG data, demonstrating their prowess in studies by Shoka et al. [7] and Zhao et al. [8].

Various classification models achieve high accuracy in seizure detection, each with distinct strengths and limitations. ML models, such as support vector machines and random forests, offer interpretability and efficiency but might struggle with capturing complex relationships [4], [9]. DL models, including CNNs and Transformers, boast remarkable effectiveness in learning intricate patterns, but face potential drawbacks of computational cost and reliance on large datasets [7], [8]. Emerging hybrid models, such as the graph-generative neural network, demonstrate promising outcomes by combining the strengths of CNNs and Transformers, achieving high accuracy with improved spatial–temporal resolution [14]. The details of various approaches are illustrated in Table I.

Overall detection accuracy consistently exceeds 95%, with some studies even reaching near-perfect scores [4], [15]. However, accurately identifying true seizures remains a primary concern, with many studies achieving sensitivities above 98% [5], [10]. Reducing false positives (FP), crucial for clinical acceptance, has sparked promising approaches utilizing real-time prediction and adaptive decision-making [10]. Real-time detection capabilities are vital for timely intervention, and studies, such as by Shen et al. [11], showcase encouraging results for practical applications.

Research efforts continue to push boundaries, exploring avenues for even more robust and impactful seizure detection systems. Personalized seizure detection, tailoring models to individual patients' specific EEG patterns, holds immense promise [10]. Expanding research beyond frequently studied seizure types toward inclusivity for diverse patient populations is another crucial area of advancement. Enhancing the interpretability and explainability of DL models is vital for garnering clinical acceptance and building trust in their predictions. Finally, rigorous testing on diverse patient populations through large-scale clinical validation is necessary before real-world implementation can ensure effectiveness and safety.

TABLE I
LITERATURE REVIEW

Author name	Year	Methodology used
Shoka et al. [7]	2023	CNN-based epileptic seizures detection framework using encrypted EEG signals.
Farooq et al. [9]	2023	ML-based epileptic seizure detection.
Zhao et al. [8]	2023	Interactive local and global feature coupling for EEG-based epileptic seizure detection.
Xu et al. [10]	2023	Real-time epileptic seizure detection via probabilistic prediction.
Amiri et al. [6]	2023	Automatic epileptic seizure detection using sparse common spatial pattern and adaptive short-time Fourier transform-based synchrosqueezing transform.
Shen et al. [11]	2023	Real-time epilepsy seizure detection based on EEG using tunable-Q wavelet transform and CNN.
Natu et al. [12]	2023	Hybrid convolutional bidirectional GRU-based model for epileptic seizure detection.
Wijayanto et al. [13]	2023	Epileptic seizure detection on a compressed EEG signal using energy measurement.

The field of automated epilepsy seizure detection using EEG signals has witnessed remarkable progress. While continuous efforts are needed to address challenges, such as personalization, FP, and clinical integration, the future holds tremendous promise for developing reliable and robust seizure detection systems. Such advancements have the potential to significantly improve patient care and quality of life for millions living with epilepsy.

The key contributions of the proposed study encompass the following.

- 1) Formulating a specialized DL architecture tailored explicitly for detecting epileptic seizures within EEG data.
- 2) Incorporating principles from fuzzy logic to augment the interpretability of the DL model.
- 3) Assessing and evaluating the performance of the proposed model using established benchmark EEG datasets.
- 4) Conducting a comparative analysis with existing state-of-the-art approaches to showcase the effectiveness and superiority of our method.
- 5) Pioneering effort in implementing explainable FDL specifically for epileptic seizure detection.
- 6) Evaluating the effectiveness of the proposed FDL by comparing it with other relevant approaches i.e., LSTM and 1D-CNN.

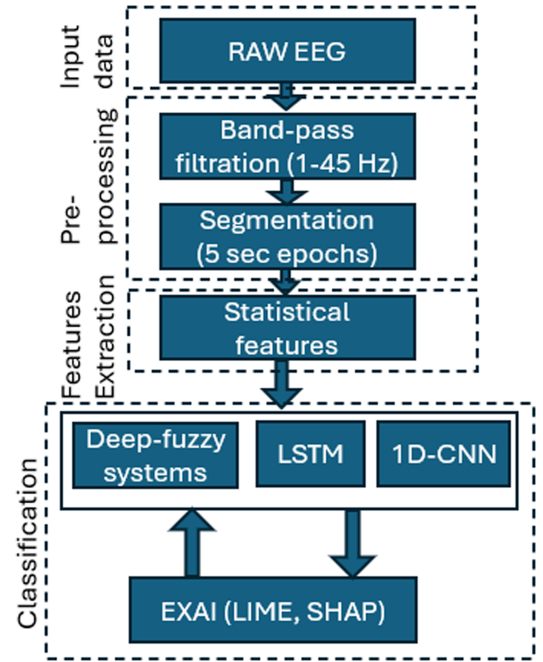


Fig. 2. Block diagram of proposed methodology.

III. METHODOLOGY

Fig. 2 displays the suggested methodology's block diagram, illustrating the seamless integration of preprocessing, feature extraction, model architecture, and XAI techniques into a comprehensive framework.

A. Dataset Description

The CHB-MIT Scalp EEG dataset, publicly accessible and documented in [16], comprises EEG recordings obtained at the Children's Hospital Boston and MIT. It encompasses recordings from pediatric subjects presenting diverse neurological conditions, such as epilepsy, brain tumors, and individuals without neurological disorders serving as healthy controls. Widely employed in EEG signal processing research, particularly in the development of seizure detection and classification methods, this dataset can be obtained from the Physionet website, a biomedical signal processing resource established by MIT. In total, the CHB-MIT Scalp EEG Dataset encompasses 23 subjects, including 21 with epilepsy and two with healthy controls. The dataset incorporates 49 EEG recordings, each with a duration ranging from 23 to 42 min. The recordings are sampled at 256 Hz and are stored in the European Data Format. Notably, the dataset encompasses both continuous EEG recordings and those specifically capturing seizure events, amounting to approximately 2500 seizures in total.

B. Preprocessing

Prior to applying DL techniques for epileptic seizure detection using EEG signals, thorough preprocessing steps are essential to ensure the quality and consistency of the data. These preprocessing steps involve the following.

Bandpass filtering: To eliminate noise and artifacts outside the relevant frequency range, noncausal bandpass filter is applied to continuous segments of raw EEG data. The filter's frequency range is set to 1–45 Hz, encompassing the spectral range of epileptic seizures. The filter design utilizes the firwin method and a windowed time-domain design strategy, as described by Pieper et al. [17]. Specifically, a Hamming window is used, which has a stopband attenuation of 53 dB and a passband ripple of 0.0194.

Segmentation: The filtered EEG signal is segmented into epochs of 5-s duration. This segmentation strategy allows for the analysis of shorter time intervals, effectively capturing potential seizure activity. In addition, it increases the dataset size, enhancing the robustness and diversity of the data for subsequent DL model training. These preprocessing steps ensure that the EEG data are clean, consistent, and well suited for DL algorithms, enabling accurate and reliable seizure detection.

C. Features Extraction

In this study, feature extraction plays a pivotal role in capturing relevant information from EEG signals, facilitating the subsequent classification of epileptic seizure events. Each EEG signal is characterized by 23 channels, and a comprehensive set of 13 features is extracted from each channel. This results in a total of 299 features per EEG signal, providing a rich representation of the signal's characteristics.

The 13 extracted features for each channel (data) include the following.

Mean:

$$\text{Mean}(\text{data}) = \frac{1}{N} \sum_{i=1}^N x_i. \quad (1)$$

Standard Deviation (std):

$$\text{std}(\text{data}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \text{Mean}(\text{data}))^2}. \quad (2)$$

Peak-to-Peak Amplitude (ptp):

$$\text{ptp}(\text{data}) = \max(\text{data}) - \min(\text{data}). \quad (3)$$

Variance (var):

$$\text{var}(\text{data}) = \frac{1}{N} \sum_{i=1}^N (x_i - \text{Mean}(\text{data}))^2. \quad (4)$$

Minimum value (minim):

$$\text{minim}(\text{data}) = \min(\text{data}). \quad (5)$$

Maximum value (maxim):

$$\text{maxim}(\text{data}) = \max(\text{data}). \quad (6)$$

Index of minimum value (argminim):

$$\text{argminim}(\text{data}) = \arg \min(\text{data}). \quad (7)$$

Index of maximum value (argmaxim):

$$\text{argmaxim}(\text{data}) = \arg \max(\text{data}). \quad (8)$$

Mean square:

$$\text{mean_square}(\text{data}) = \frac{1}{N} \sum_{i=1}^N x_i^2. \quad (9)$$

Root mean square (RMS):

$$\text{RMS}(\text{data}) = \sqrt{\text{mean_square}(\text{data})}. \quad (10)$$

Absolute differences signal (abs_diffs_signal):

$$\text{abs_diffs_signal}(\text{data}) = \sum_{i=1}^N |x_i - x_{i-1}|. \quad (11)$$

Skewness:

$$\text{Skewness}(\text{data}) = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \text{Mean}(\text{data}))^3}{\left(\frac{1}{N} \sum_{i=1}^N (x_i - \text{Mean}(\text{data}))^2\right)^{\frac{3}{2}}}. \quad (12)$$

Kurtosis:

$$\text{Kurtosis}(\text{data}) = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \text{Mean}(\text{data}))^4}{\left(\frac{1}{N} \sum_{i=1}^N (x_i - \text{Mean}(\text{data}))^2\right)^2}. \quad (13)$$

These features collectively offer a comprehensive characterization of the EEG signals, capturing statistical, amplitude, and shape-related properties. The resulting feature set serves as a foundation for subsequent ML models, providing discriminative information essential for accurate seizure detection and classification.

D. Fuzzy DL

Fuzzy logic systems are a mathematical framework that deals with reasoning and decision-making in situations where imprecision and uncertainty exist [18]. Unlike classical binary logic, where a statement can only be true or false, fuzzy logic allows for the representation of partial truth. It finds applications in various fields, such as control systems, artificial intelligence, and decision support systems. In a fuzzy logic system, variables are not assigned a definite truth value (0 or 1) but a degree of membership in a fuzzy set between 0 and 1. The degree of membership indicates the extent to which an element belongs to a set. The fundamental operations in fuzzy logic include fuzzification, rule evaluation, and defuzzification.

Fuzzification:

$$\mu_A(x) = \frac{1}{1 + \left(\frac{x-c}{a}\right)^{2b}} \quad (14)$$

where a , b , and c are parameters determining the shape of the fuzzy membership function.

Rule evaluation:

$$\text{Rule}_i : \text{IF (condition)}_i \text{ THEN (consequence)}_i. \quad (15)$$

Defuzzification:

$$\text{Output} = \frac{\sum (\text{Rule Output} \times \text{Degree of Activation})}{\sum \text{Degree of Activation}}. \quad (16)$$

Proposed model architecture—Base model: The foundational aspect of our architecture is the base model, designed for feature

extraction from EEG signals. It comprises three dense layers, each contributing to the extraction of distinctive features. These layers consist of 128, 32, and 32 units, incorporating rectified linear unit (ReLU) activation functions to introduce nonlinearity and enhance the model's ability to capture complex patterns in EEG data.

Fuzzy layers: To augment the model's effectiveness in capturing nuanced patterns indicative of epileptic seizures, custom fuzzy layer instances are sequentially applied. These fuzzy layers introduce fuzzy logic principles, contributing to improved seizure detection. The number of fuzzy sets varies in each layer, with 64 for "fuzzy1," 32 for "fuzzy2," and 16 for "fuzzy3."

The number of fuzzy sets in each layer was determined through empirical analysis and experimentation. We conducted a series of tests to identify the optimal configuration that provided the best balance between model performance and interpretability. This approach ensures that the model captures the complex temporal and spatial dynamics of seizure-related EEG patterns. For further understanding, readers may refer to foundational works on fuzzy logic applications in signal processing, such as Ross [19] and Güler and Übeyli [20].

The membership function used in each fuzzy layer is defined as follows:

$$\text{Membership function} = \exp \left(- (\text{inputs} - \text{mf})^2 \right). \quad (17)$$

Here, the following holds.

- 1) $\exp$ represents the exponential function.
- 2) inputs is the input to the membership function.
- 3) mf is the membership function parameter (weight) learned during training.
- 4) $(\text{inputs} - \text{mf})^2$ calculates the squared difference between the input and the membership function parameter.

This formula computes the degree of membership or "closeness" of the input to the fuzzy set represented by the specific membership function. The fuzzy rules are implicitly formed through the combination of these membership functions and fuzzy inference operations.

Output layer: The final layer is a dense layer with a single unit and utilizes a sigmoid activation function, tailored for binary classification. This layer is crucial for providing the ultimate prediction regarding the presence or absence of an epileptic seizure in EEG data. The architecture is shown in Fig. 3.

Key adjustments and improvements—Activation functions: ReLU activation functions in the base model's dense layers contribute significantly to capturing relevant EEG signal features, aiding in improved detection accuracy.

Fuzzy layers: The incorporation of exponential membership functions in fuzzy layers enhances the model's ability to discern subtle patterns associated with seizure events. The choice of the number of fuzzy sets in each layer is based on experimental tuning to balance complexity and interpretability. For instance, the first fuzzy layer uses 64 fuzzy sets to capture a broad range of patterns, the second layer refines these patterns with 32 fuzzy sets, and the third layer further fine-tunes the analysis with 16 fuzzy sets. This hierarchical structuring of fuzzy sets aligns with

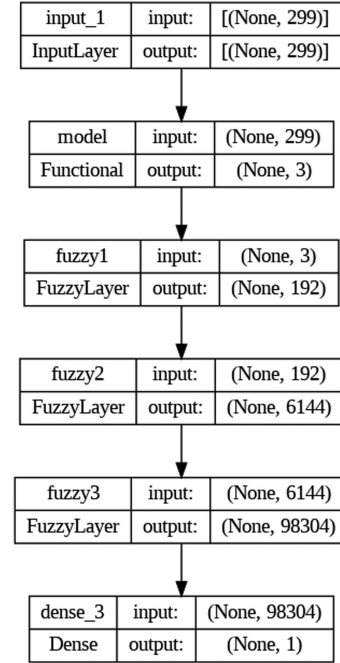


Fig. 3. FDL architecture.

the multilevel nature of EEG signal processing, enhancing the system's overall interpretability.

Output layer: Sigmoid activation facilitates precise binary classification, allowing the model to make accurate predictions regarding the occurrence of epileptic seizures.

This architecture was instrumental in our research on epileptic seizure detection using EEG signals. The nuanced combination of traditional feature extraction with fuzzy logic principles led to substantial improvements in detection accuracy. The configuration of fuzzy sets in each layer was chosen based on experimental evidence, ensuring the model effectively captures the complex patterns in EEG data while remaining interpretable.

Interpretability mechanisms: While Shapley additive explanations (SHAP) and local interpretable model-agnostic explanations (LIME) are utilized to explain the contribution and importance of the statistical features fed into the model, the interpretability of the fuzzy logic system itself is evident in its structure and the nature of fuzzy rules applied. The fuzzy layers transform the input data into a space where each dimension corresponds to a degree of membership to certain fuzzy sets, inherently providing a human-understandable framework for the model's decision-making process. This approach leverages both the transparency of fuzzy logic and the power of DL, ensuring that the model's predictions can be interpreted in a meaningful way.

The configuration of fuzzy sets in each layer was carefully chosen based on empirical analysis and the complex nature of EEG signals during seizures. Through a series of experiments, we identified the optimal number of fuzzy rules that provided the best performance in terms of classification accuracy and interpretability.

Algorithm 1: Training Algorithm for FDL Model.

```

1: Initialize model parameters
2: while not converged do
3:   Obtain EEG signal segment
4:   Preprocess signal (filtering and segmentation)
5:   Extract features
6:   Fuzzification using custom fuzzy layers
7:   Perform forward pass through dense layers
8:   Compute binary cross-entropy loss
9:   Backpropagate and update model parameters
10: end while
11: Return trained model

```

E. LSTM Networks

To evaluate the performance of different models for epileptic seizure detection, we deployed an LSTM network independently. LSTM networks are well suited for capturing temporal dependencies within EEG signals [21], making them appropriate for time series analysis, such as this.

The LSTM architecture developed for our task consists of the following: several LSTM layers to capture the temporal characteristics of EEG signals, and dense layers after the LSTM layers to further process the extracted features. The initial dense layer employs the hyperbolic tangent (tanh) activation function, while the final dense layer uses the sigmoid activation function for binary classification. Regularization methods, such as dropout layers, were part of the early design but were commented out in the final model. They can be reactivated by uncommenting the relevant lines in the code.

The model was compiled using the binary crossentropy loss function and the Adam optimizer, ensuring efficient training. By evaluating this LSTM model independently, we aim to benchmark its performance against the proposed FDL model.

F. 1D-Convolutional Neural Network

In addition to the LSTM model, we also independently deployed a 1D-CNN to analyze the performance of different architectures for epileptic seizure detection. The 1D-CNN is adept at capturing local patterns in sequential data, making it a suitable choice for processing EEG signals [22].

The 1D-CNN architecture for our task includes the following.

1) *Convolutional layers:*

- a) A convolutional layer with 32 filters, each with a kernel size of 3 and ReLU activation function, followed by a max pooling layer with a pool size of 2.
- b) Next, a second convolutional layer is added, which includes 64 filters, each of size 3, and makes use of the ReLU activation function. This is then followed by a max pooling layer with a pool size of 2.
- c) A third convolutional layer containing 128 filters, each with a kernel size of 3, utilizing the ReLU activation function, and succeeded by a max pooling layer with a pool size of 2.
- 2) A flattening layer to convert the 3-D feature maps into a 1-D feature vector.

3) *Dense layers:*

- a) A dense layer with 64 units and ReLU activation function.
- b) A dropout layer with a dropout rate of 0.5 for regularization.
- c) A final dense layer with 1 unit and sigmoid activation function for binary classification.

The convolutional operation in each layer can be represented as follows:

$$\text{Convolution: } y_i = \sum_{j=0}^{m-1} w_j \cdot x_{i+j} + b \quad (18)$$

where y_i represents the output of the convolutional layer at position i , w_j are the filter weights, x_{i+j} are the input values within the receptive field of the filter, and b is the bias term. The model is compiled using the binary crossentropy loss function and the Adam optimizer.

By independently deploying the 1D-CNN model, we aim to compare its performance with the proposed FDL model. This comparative analysis allows us to determine the most effective model for accurate epileptic seizure detection.

G. Explainable AI

XAI refers to the set of techniques and methods designed to enhance the transparency and interpretability of AI models. As AI systems, particularly DL models, become increasingly complex and powerful, understanding their decision-making processes becomes crucial for user trust, ethical considerations, and deployment in sensitive domains. XAI aims to demystify closed-box models by providing insights into how and why a model arrives at a specific decision or prediction [23]. Techniques, such as LIME and SHAP, are commonly used in XAI to generate human-understandable explanations for AI model outputs.

1) *Local Interpretable Model-Agnostic Explanations:* LIME is a model-agnostic explanation technique that can be used to explain any ML model. It works by fitting a local interpretable model to the original model's predictions in the vicinity of the instance to be explained. The LIME is typically a simple model, such as a linear regression model, which can be easily interpreted [24]

$$f_{\text{LIME}}(x) = f_{\text{local}}(x) = \sum_{i=1}^n \alpha_i f(x - \epsilon_i e_i) \quad (19)$$

where the following holds.

- 1) $f_{\text{LIME}}(x)$ is the prediction of the LIME for the instance x .
- 2) $f_{\text{local}}(x)$ is the prediction of the original model for the instance x , locally approximated by the LIME.

The pseudocode for LIME is as follows.

2) *Shapley Additive Explanations:* SHAP is a model-specific explanation technique that uses Shapley values to explain the predictions of an ML model. Shapley values are a cooperative game theory concept that assigns each feature in an instance

Algorithm 2: Working of LIME for the Proposed Work.**Input:**

- EEG signal, $\mathbf{X}$
- Feature extraction, FE
- XAI model, XAI
- Instance to be explained, $\mathbf{X}_{\text{instance}}$

Output: Local interpretable model (LIME), LIMXai**Algorithm:**

- 1) Extract features: $\mathbf{F} = \text{FE}(\mathbf{X})$
- 2) Generate perturbed instances:
 $\mathbf{X}_{\text{perturbed}} = \text{Perturb}(\mathbf{X}_{\text{instance}})$
- 3) Get predictions: $\mathbf{Y}_{\text{perturbed}} = \text{XAI}(\mathbf{X}_{\text{perturbed}})$
- 4) Train LIME:
 $\text{LIMXai} = \text{Train}(\mathbf{X}_{\text{perturbed}}, \mathbf{Y}_{\text{perturbed}})$
- 5) Explain: $\text{ExplanationXAI} = \text{LIMXai}(\mathbf{X}_{\text{instance}})$

TABLE II
COMPARISON OF LIME AND SHAP

Feature	LIME	SHAP
Model-agnostic	Yes	No
Local versus global explanations	Local	Global
Computational complexity	Lower	Higher
Theoretical properties	Weaker	Stronger

Algorithm 3: Working of SHAP for the Proposed Work.**Input:**

- EEG signal, $\mathbf{X}$
- Feature extraction, FE
- XAI model, XAI
- Instance to be explained, $\mathbf{X}_{\text{instance}}$

Output: SHAP values, ΦSHAP **Algorithm:**

- 1) Extract features: $\mathbf{F} = \text{FE}(\mathbf{X})$
- 2) Calculate Shapley values:
 $\Phi\text{SHAP} = \text{Shapley}(\mathbf{X}_{\text{instance}}, \text{XAI})$
- 3) Explain: $\text{Explanation XAI} = \text{Explain}(\Phi\text{SHAP})$

a value that represents its contribution to the model's prediction [25]

$$f_{\text{SHAP}}(x) = \sum_{j=1}^m \phi_j(x) f_j(x) \quad (20)$$

where the following holds.

- 1) $f_{\text{SHAP}}(x)$ is the prediction of the SHAP model for the instance x .
- 2) $\phi_j(x)$ is the Shapley value for feature j in the instance x .
- 3) $f_j(x)$ is the value of feature j in the instance x .

The working of LIME in the proposed framework is as follows.

3) *Comparison of LIME and SHAP:* The Table II summarizes the key differences between LIME and SHAP. Overall, LIME is a good choice for explaining individual predictions, while SHAP is a good choice for understanding the overall behavior of a model.

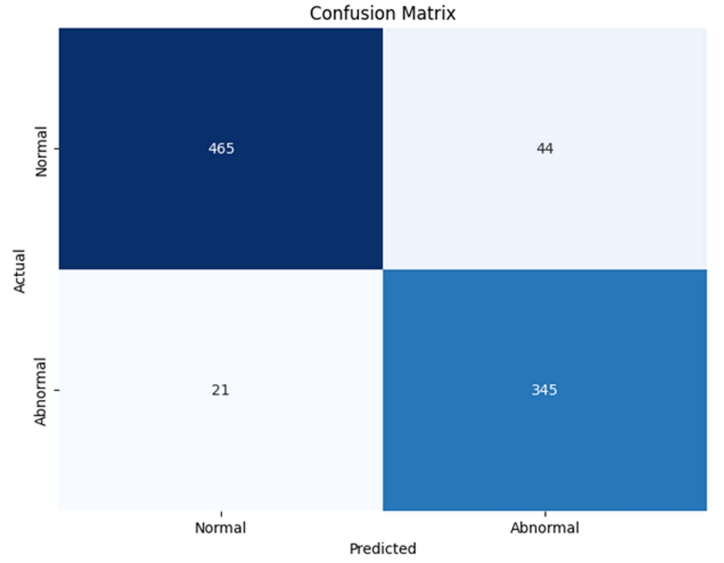


Fig. 4. Confusion matrix.

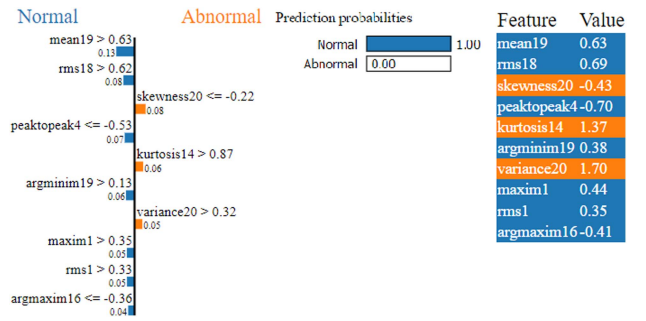


Fig. 5. LIME explanation: Interactive feature contributions with table for normal signal.

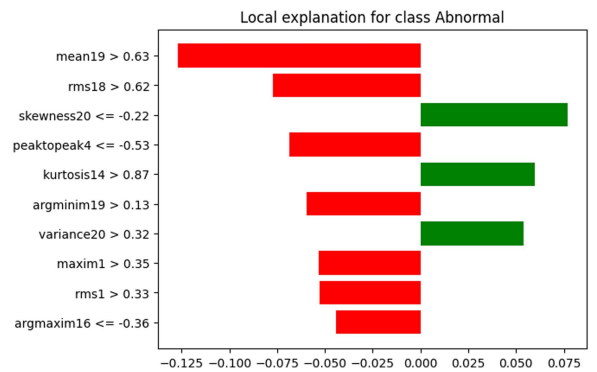


Fig. 6. LIME explanation: Feature contributions bar chart for normal signal.

IV. RESULTS AND DISCUSSION

In this study, we aimed to evaluate the effectiveness of three distinct techniques—1D-CNN, LSTM, and FDL—in detecting epileptic seizures from EEG signals, with the assistance of XAI to enhance the interpretability and performance of the models.

TABLE III
COMPARISON OF THE RESULTS ACHIEVED BY DIFFERENT APPROACHES

Method	Accuracy	Precision	Recall	F1-Score	Precision	Recall	F1-Score
		Normal			Seizure		
1D-CNN	0.87	0.95	0.81	0.88	0.80	0.95	0.87
LSTM	0.89	0.90	0.89	0.90	0.88	0.89	0.89
FDL	0.92	0.96	0.91	0.93	0.89	0.94	0.91

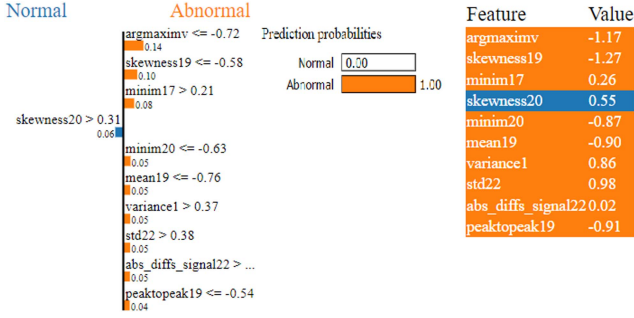


Fig. 7. LIME explanation: Interactive feature contributions with table for abnormal signal.

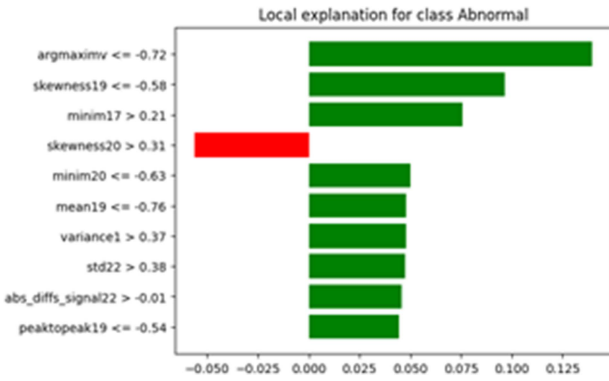


Fig. 8. LIME Explanation: Feature contributions bar chart for abnormal signal.

The findings presented in this article originate from a comprehensive methodology that seamlessly integrates preprocessing, feature extraction, model architecture, and XAI techniques.

The dataset used in this study consists of EEG recordings categorized into “normal” and “abnormal” classes. The dataset was divided into training, validation, and test sets with a ratio of 70:15:15, respectively. This division ensures that the models are trained on a diverse set of data and validated and tested on unseen samples to assess their generalizability.

The EEG signals underwent preprocessing, employing a 1–45 Hz bandpass filter aimed at enhancing signal specificity for seizure detection. Feature extraction involved computing 299 features per signal, encompassing statistical measures, such as mean, standard deviation, and skewness, among others, detailed in Section III-C. The models were configured with specific settings as detailed in Sections III-D–III-F to optimize their performance.

Moreover, to enhance model interpretability, two XAI techniques, namely, LIME and SHAP, are utilized. LIME furnishes local interpretability by generating explanations for individual predictions, providing insights into the model’s decision-making on a case-by-case basis. Simultaneously, SHAP analysis contributes to global interpretability by assigning Shapley values to each feature, elucidating complex relationships and dependencies within the dataset. This comprehensive approach, spanning preprocessing, feature extraction, model architecture, and XAI, ensures a nuanced comprehension of the FDL model’s performance in EEG signal classification and furnishes valuable insights into the influential features driving its predictions

$$\text{Precision} = \frac{TP}{TP + FP} \quad (21)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (22)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (23)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (24)$$

The precision, recall, accuracy, and F1-score metrics as in (21)–(24), respectively, which are using true positive (TP), true negative (TN), FP, and false negative (FN) values, offer a comprehensive evaluation of the model’s performance. The detail results for each approach are illustrated in the Table III. The 1D-CNN model demonstrated promising performance, achieving an accuracy of 87.20%. While the model exhibited high precision (0.95) for normal instances and seizure (abnormal) instances (0.80), its recall for normal instances (0.81) was relatively lower compared with the seizure instances (0.95). Consequently, the model achieved an F1-score of 0.88 for normal instances and 0.87 for seizure instances. Despite its balanced F1-scores across classes, the lower recall for normal instances suggests potential missed detections of normal EEG patterns as seizures.

Similarly, the LSTM model showcased robust performance, with an accuracy of 89.14%. With precision values of 0.90 for normal instances and 0.88 for seizure instances, and recall values of 0.89 for both classes, the LSTM model demonstrated consistent performance in identifying instances of both normal and seizure (abnormal) classes. This led to balanced F1-scores of 0.90 for normal instances and 0.89 for seizure instances, reflecting the model’s ability to accurately classify instances from EEG signals.

FDL emerged as the top-performing technique, achieving an accuracy of 92.57%. The model exhibited high precision values of 0.96 for normal instances and 0.89 for seizure instances,

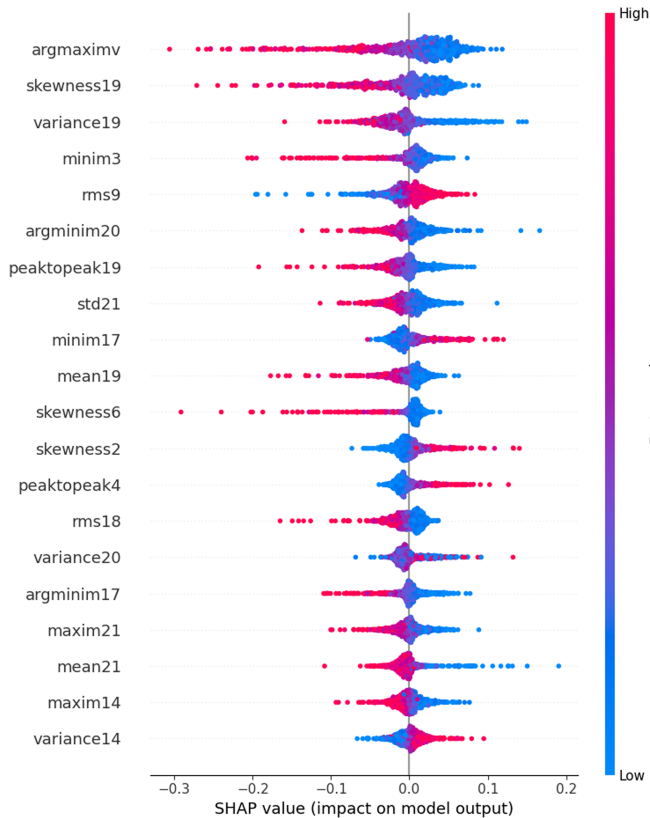


Fig. 9. SHAP-based explainability (dependency).

coupled with recall values of 0.91 for normal instances and 0.94 for seizure instances. Consequently, the F1-scores were 0.93 for normal instances and 0.91 for seizure instances. These results indicate the model's ability to effectively minimize FP while capturing TP, leading to superior performance in seizure detection. The confusion matrix for the FDL-based approach is shown in Fig. 4.

In the pursuit of enhancing model interpretability, two XAI techniques, LIME and SHAP, were employed. LIME was leveraged to generate local explanations for individual predictions made by the FDL model. The top ten features contributing to predictions for each class were identified and visually presented in Figs. 5–8. These features provide valuable insights into the specific characteristics of EEG signals influencing the model's decision-making at a local level. LIME's focus on local interpretability allows practitioners to gain an understanding of the model's reasoning for individual instances.

SHAP analysis was employed to uncover the global interpretability of the FDL model. By attributing Shapley values to each feature, SHAP facilitates the identification of dependencies and correlations between the extracted statistical features, as illustrated in Figs. 9 and 10. The Shapley values offer nuanced insights into the impact of individual features on the model's output, contributing to a holistic understanding of feature importance at a global level. SHAP's capacity for global interpretability enables practitioners to comprehend the broader patterns and relationships within the dataset. Interpreting the plot involves understanding which features have the most substantial

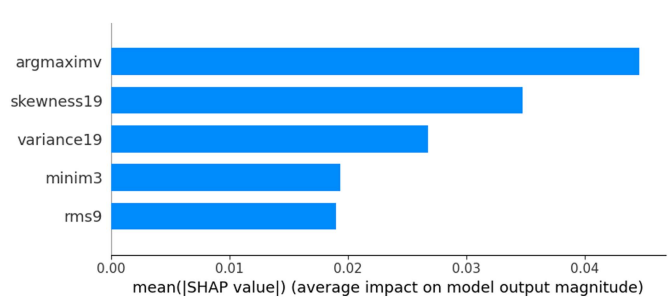


Fig. 10. SHAP-based explainability.

impact on pushing the model output toward the positive or negative direction. Features with consistently high positive Shapley values across instances contribute positively to the model output, while those with consistently high negative values contribute negatively. This aids in identifying the key drivers influencing the model's decision-making process and provides insights into feature importance for the given dataset.

V. CONCLUSION

Effectively addressing the challenges presented by the unpredictable and recurring nature of epileptic seizures, particularly within high patient flow settings, remains a critical priority. Timely identification of these seizures is fundamental to ensuring optimal patient care. The proposed custom FDL-based architecture has demonstrated notable precision (0.96 for “normal” and 0.89 for “abnormal”) alongside robust recall rates (0.91 for “normal” and 0.94 for “abnormal”). The comparison with LSTM and 1D-CNN models validated the effectiveness of our proposed FDL approach. Incorporating LIME and SHAP has enhanced interpretability, providing valuable insights into the pivotal features influencing predictions. This comprehensive approach, spanning preprocessing, feature extraction, and the utilization of XAI techniques, has constituted a cohesive strategy for achieving both accuracy and interpretability in seizure detection. The proposed work has significantly contributed to the advancement of EEG-based diagnostics, underscoring the practical significance of this methodology within clinical neurology.

As part of our ongoing research, future work will focus on extending the applicability of our methodology to neonatal EEG datasets for early epileptic seizure detection using XAI.

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